

Collection of Waste:

Collection of separated or unseparated solid waste in urban areas is difficult and complex because of the generation of residential, commercial or industrial solid waste takes place in every home, apartment, commercial or industrial facility, streets, parks and even in vacant areas.

❖ Types of Collection Service

The Various types of collection service are

1. Municipal Collection service / Residential collection service
 - A. From low rise detached dwellings
 - B. From low and medium rise apartment
 - C. From high rise apartment
2. Commercial and Industrial collection service

1. Municipal Collection service / Residential collection service

Collection service varies depending upon the type of dwelling unit, collection for low rise detached dwellings and collection for medium and high rise apartments are considered separately.

The most common types of residential services in various parts of the country include

1. Curb
2. Alley
3. Set out – Set back
4. Set out
5. Backyard carrying

1. **Curb** – is used for low rise detached dwelling. This is a manual type of collection system where in the waste are collected in a curb on a collection day and the containers are returned back to their storage location until the next collection.

2. **Alley** - are part of the basic layout of a city or a given residential area. Alleys are storage of container used for solid waste collection.
3. **Set out – Set back** – Containers are set out from the owners property and set back after being emptied by additional crew.
4. **Set out** – this service is essentially the same as set-out & set-back, except that the home owner is responsible for returning the container back their storage location.
5. **Backyard Carrying** – The collection crew is responsible for entering the owners property and removing the waste from their storage location.

2. Commercial and Industrial collection service

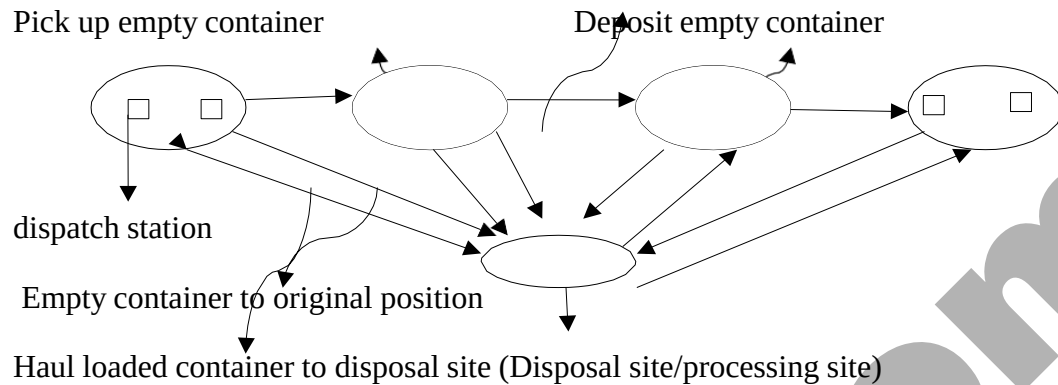
- Both manual and mechanical means are used to collect waste from commercial and industrial areas. To avoid traffic congestion during the day time, solid waste from commercial establishment in many cities are collected in the late evening and early morning.
- Where manual collection is used, Wastes are put in plastic bags, cardboard boxes and other disposable containers that are placed at curbs for collection.
- The Collection services provided to large apartment buildings, residential complexes, commercial and industrial areas are provided to these centers on the use of movable containers, stationery containers, and large stationery compactors.
- Compactors are of the type that can be used to compress materials into large containers.

❖ Types of Collection Systems

Based on the mode of operation, collection systems are classified in to two categories:

1. Hauled container systems
2. Stationery container systems

1. Hauled Container system: Drive to next loaded container



Collection system in which the containers used for the storage of waste are hauled to the processing, transfer or disposal site, emptied and returned to either their original location or some other location are defined as hauled container system. The collector is responsible for driving the vehicle, loading the containers and emptying the contents of containers at disposal site.

There are two main types of hauled container system

1. Tilt Frame container
2. Trash-trailors

Systems that use tilt frame loaded vehicles often called drop boxes are ideally suited for collection of all types of solid waste and rubbish from locations where generation rates warrants the use of large containers. Because of large volume that can be hauled, the use of tilt frame hauled container systems has become widespread, especially among private services.

Trash trailers are better for collection of especially heavy rubbish such as sand, timber and metal scrap and often are used for collection of demolition waste at construction sites.

Advantages

1. Hauled container systems are flexible containers of many different shapes and sizes are available for collection of all types of wastes.

Disadvantages

1. Because the container used in this system must be filled, usually the use

of verylarge containers often lead to low volume utilization.

2. Through this system requires only one truck and driver to accomplish the collection cycle, each container picked up requires a round trip to disposal site.

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2. Stationery Collection system

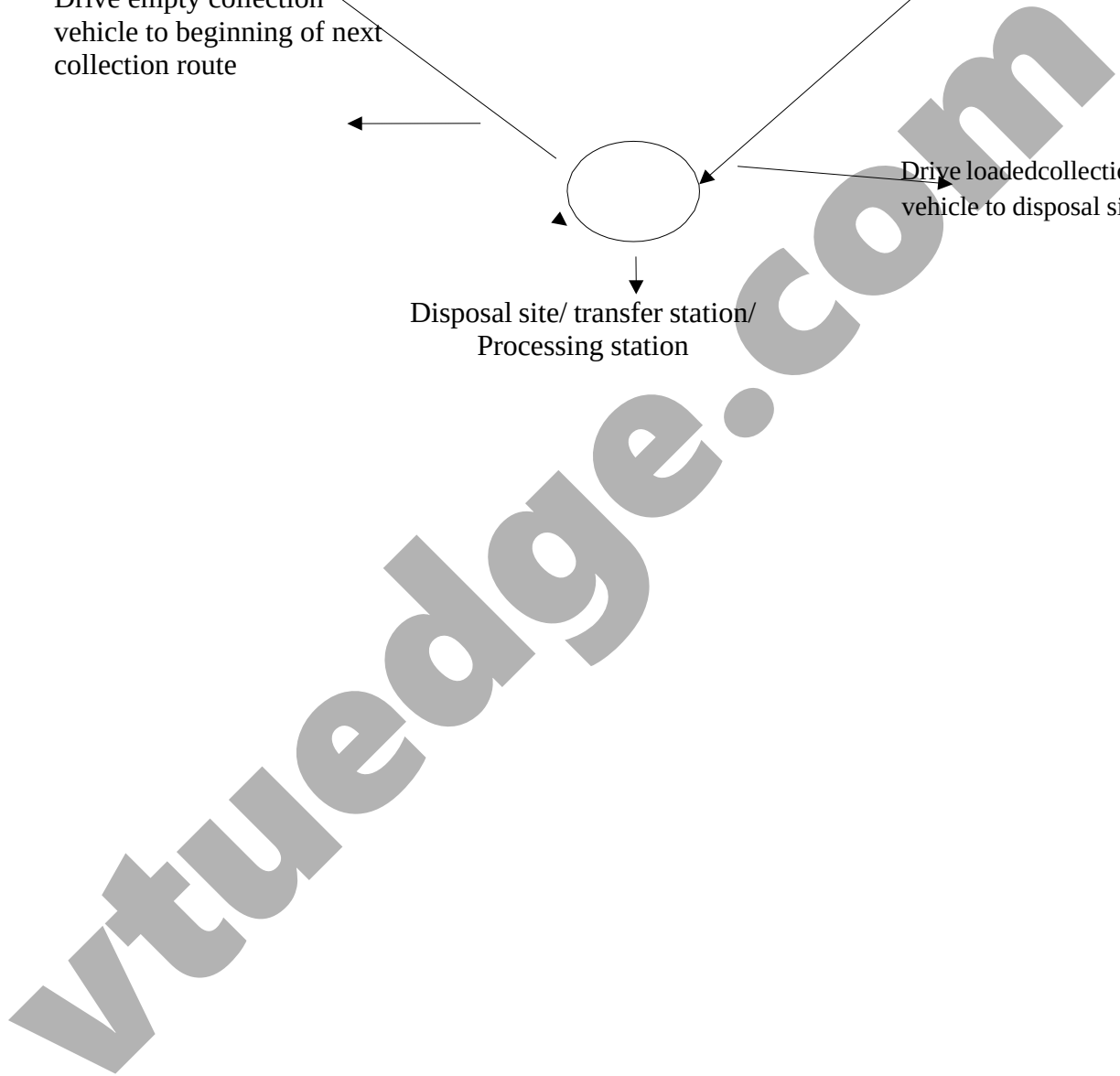
Drive to next pick up point

load contents from container into
collection vehicle

Drive empty collection
vehicle to beginning of next
collection route

Drive loaded collection
vehicle to disposal site

Disposal site/ transfer station/
Processing station



Waste Management (module 3)

. Collection system in which the container used for the storage of waste remains at the point of waste generation, except when moved for collection are defined as stationery container system.

There are two main types of stationery container systems

1. Those in which self loading compactors used
2. Those in which manually loaded vehicles are used.

Trips to the disposal site, transfer station, or processing station are made after the contents of a no of containers have been collected and compacted and collection vehicle is full. Because a variety of container sizes and types are available, these systems may be used for collection of all types of wastes.

Factors affecting collection system

Components	Factors to Consider	
Crew size	• labour cost • distance between containers • size and types of containers • loading accessories available in the truck • collection vehicle used	
Container type	• solid wastes generation rate • density of waste generation • street width • traffic volume • collection crew configuration • standard of living	
Collection accessory	• labour cost • protection of worker's health	
Vehicle size/type	• street width, traffic volume • solid waste generation rates • crew size • viability of a transfer station	
Collection route	• street width, traffic volume • direction of traffic flow • solid waste generation rates • spatial distribution of wastes • local topography	
Transfer station	• distance between disposal site and collection area • hauling cost for small and large trucks • cost of transferring the solid wastes from small to large trucks	

Storage containers:

Proper container selection can save collection energy, increase the speed of collection and reduce crew size. Most importantly, containers should be functional for the amount and type of materials and collection vehicles used. Containers should also be durable, easy to handle, and economical, as well as resistant to corrosion, weather and animals. In residential areas, where refuse is collected manually, standardised metal or plastic containers are typically required for waste storage. When mechanised collection systems are used, containers are specifically designed to fit the truck-mounted loading mechanisms. While evaluating residential waste containers, consider the following:

- **efficiency**, i.e., the containers should help maximise the overall collection efficiency.
- **convenience**, i.e., the containers must be easily manageable both for residents and collection crew.
- **compatibility**, i.e., the containers must be compatible with collection equipment.
- **public health and safety**, i.e., the containers should be securely covered and stored.
- **ownership**, i.e., the municipal ownership must guarantee compatibility with collection equipment.

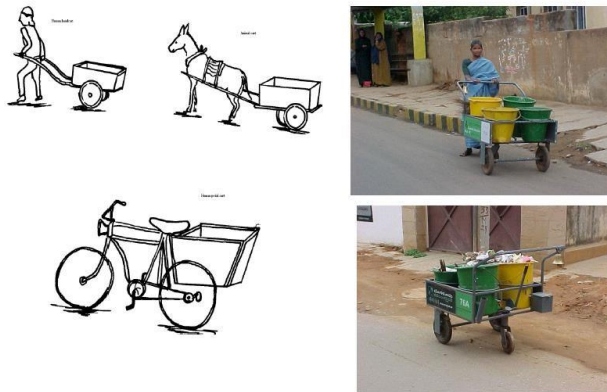
COLLECTION VEHICLES

Almost all collections are based on collector and collection crew, which move through the collection service area with a vehicle for collecting the waste material. The collection vehicle selected must be appropriate to the terrain, type and density of waste generation points, the way it travels and type and kind of material. It also depends upon strength, stature and capability of the crew that will work with it. The collection vehicle may be small and simple (e.g., two-wheeled cart pulled by an individual) or large, complex and energy intensive (e.g., rear loading compactor truck). The most commonly used collection vehicle is the dump truck fitted with a hydraulic lifting mechanism. A description of some vehicle types follows: (i) Small-

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scale collection and muscle-powered vehicles: These are common vehicles used for waste collection in many countries and are generally used in rural hilly areas. As Figure illustrates, these can be small rickshaws, carts or wagons pulled by people or animals, and are less expensive, easier to build and maintain compared to other vehicles.

Small-scale Collection Vehicles: An Illustration



They are suitable for densely populated areas with narrow lanes, and squatter settlements, where there is relatively low volume of waste generated. Some drawbacks of these collection vehicles include limited travel range of the vehicles and weather exposure that affect humans and animals.

(ii) **Non-compactor trucks:** Non-compactor trucks are efficient and cost effective in small cities and in areas where wastes tend to be very dense and have little potential for compaction. Figure illustrates a non- compactor truck:

Non-compactor Trucks



When these trucks are used for waste collection, they need a dumping system to easily discharge the waste. It is generally required to cover the trucks in order to prevent residue flying off or rain soaking the wastes. Trucks with capacities of 10 – 12 m³ are effective, if the distance between the disposal site and the collection area is less than 15 km. If the distance is longer, a potential transfer station closer than 10 km from the collection area is required. Non-compactor trucks are generally used, when labour cost is high. Controlling and operating cost is a deciding factor, when collection routes are long and relatively sparsely populated.

(iii) **Compactor truck:** Compaction vehicles are more common these days, generally having capacities of 12 – 15 m³ due to limitations imposed by narrow roads. Although the capacity of a compaction vehicle, illustrated in Figure, is similar to that of a dump truck, the weight of solid wastes collected per trip is 2 to 2.5 times larger since the wastes are hydraulically compacted.

Compactor Truck



A compactor truck allows waste containers to be emptied into the vehicle from the rear, front or sides and inhibits vectors (of disease) from reaching the waste during collection and transport. It works poorly when waste stream is very dense, wet, collected materials are gritty or abrasive, or when the roads are dusty. The advantages of the compactor collection vehicle include the following

- containers are uniform, large, covered and relatively visually inoffensive;
- waste is set out in containers so that the crew can pick them up quickly;
- health risk to the collectors and odour on the streets are minimised;
- Waste is relatively inaccessible to the waste pickers.

2. Transfer Means and Methods (Transport Methods)

Motor vehicles, rail road's and ocean going vessels are the principle means used to transport waste. Pneumatic and hydraulic systems have also been used.

1. Motor Vehicle Transport - Motor vehicles used to transport solid wastes on highway should satisfy the following requirement.

- The vehicles must transport wastes at minimum costs
- Wastes must be covered during haul operation

- Vehicles must be designed for highway traffic
- Vehicle capacity must be such that allowable weight limits are not exceeded
- Methods used for unloading must be simple and dependable.

In the recent years because of their simplicity and dependability, open top trailers and semitrailers have found wide acceptance for the transport of waste. Some trailers are equipped with sumps to collect any liquid that accumulate from the solid waste.

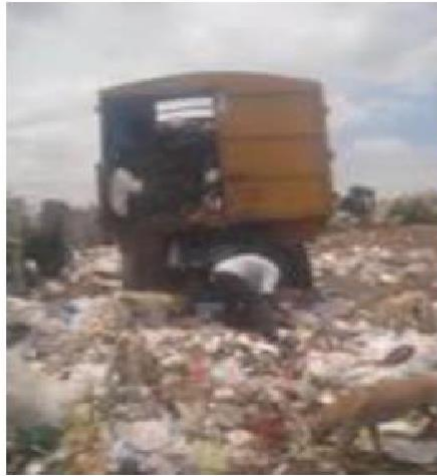
2. Rail Road Transport - Although Rail road's were commonly used for the transport of solid waste in the past, they are now used by a few communities. However renewed interest is again developing in the use of rail road for hauling solid waste especially to remote areas where highway travel is difficult and rail lines now exist.

3. Water Transport – Barges, scows and special boats have been used in the past to transport solid wastes to processing location and to seashores and ocean disposal sites, but ocean disposal is no longer practiced by developing countries.

4. Pneumatic Transport – Both low pressure air and vacuum conduit transport systems are used to transport solid waste.

- The most common application is the transport of waste from high density apartment or commercial activities to central location for processing or for loading into transport vehicles.

Transfer Station



Transfer station

A transfer station is an intermediate station between final disposal option and collection point in order to increase the efficiency of the system, as collection vehicles and crew remain closer to routes. If the disposal site is far from the collection area, it is justifiable to have a transfer station, where smaller collection vehicles transfer their loads to larger vehicles, which then haul the waste long distances. In some instances, the transfer station serves as a preprocessing point, where wastes are dewatered, scooped or compressed. A centralised sorting and recovery of recyclable materials are also carried out at transfer stations (EPA, 1989). The unit cost of hauling solid wastes from a collection area to a transfer station and then to a disposal site decreases, as the size of the collection vehicle increases. This is due to various reasons such as the following:

- labour costs remain constant
- Types of waste received.
- Processes required in recovering material from wastes.
- Required capacity and amount of waste storage desired.
- Types of collection vehicles using the facility.
- Types of transfer vehicles that can be accommodated at the disposal facilities.
- Site topography and access

Important factors that must be considered in the design of transfer station include:

1. Type of transfer operation
2. Capacity Requirement
3. Equipment and accessory requirement
4. Environmental requirement

1. Type of transfer operation : Depending on the method to load the transport vehicles, transfer stations may be classified into three types:

- A. Direct Discharge
- B. Storage Discharge
- C. Combined direct and storage discharge.

A. Direct discharge – In a direct discharge transfer station, wastes from the collection vehicles usually are emptied directly into the vehicle to be used to transport them to a place of final disposal.

- To accomplish this, these stations are constructed in a two level arrangement. The unloading platform from which the wastes from collection vehicles are discharged into the transport trailers are elevated. Direct discharge transfer stations employing stationery compactors are popular.

B. Storage discharge – In this transfer station, wastes are emptied either into a storage pit or onto a platform from which they are loaded into transfer vehicles by various types of auxiliary equipment.

- In a storage discharge transfer station, the various processing techniques like shredding, separation, magnetic separation of ferrous scraps, compaction are employed by using various types of auxiliary equipment.

C. Combined direct and storage discharge – In some transfer station, both the methods are used. Usually these are multipurpose facilities designed to service a broader range of users than a single purpose facility.

2. Capacity Requirement – The operational capacity of a transfer station must

be such that the collection vehicles do not have to wait too long to unload.

- In most cases it will not be cost effective to design the station to handle the ultimate peak no of hourly loads. An economic analysis should be made between the annual cost for the time spent by the collection vehicle waiting to unload against the incremental annual cost of a transfer station or the use of more transport equipment.
- Because of the increased cost of transport equipment, trade off analysis must also be made between the capacity of transfer station and the cost of the transport operation including both equipment and labor components.

3. Equipment and Accessory Requirements - The types and amounts of equipment required vary with the capacity of a station and its function in the waste management system.

- Specifically scales should be provided at all medium and large transfer station both to monitor the operation and to develop meaningful management and engineering data.

4. Environmental Requirements – Most of the large transfer stations are enclosed and are constructed of materials that can be maintained and cleaned easily.

- For direct discharge transfer stations with open loading areas, special attention must be given to the problem of blowing papers. Wind screens or other barriers are to be used.
- Regardless of the type of transfer station, the design and construction should be such that all accessible areas where rubbish or paper can accumulate or eliminated.

Factors affecting the location of transfer station

Whenever possible, transfer station should be located

1. As near as possible, the weighted center of the individual solid waste production areas to be served
2. Within easy access of major arterial highway routes as well as near secondary or Supplemental means of transportation.
3. Where there will be minimum public and environmental objections to the transfer operations.
4. Where construction and operation will be economical
5. Additionally if the transfer station site is to be used for processing techniques involving materials recovery and energy production, the requirement for those operation must be considered.

Types of Transfer Station

Depending on the size, transfer stations can be either of the following two types:

(i) Small to medium transfer stations:

- These are direct-discharge stations that provide no intermediate waste storage area.
- The capacities are generally small (less than 100 tonnes/day) and medium (100 to 500 tonnes/day).
- Depending on weather, site aesthetics and environmental concerns, transfer operations of this size may be located either indoor or outdoor.
- More complex small transfer stations are usually attended during hours of operation and may include some simple waste and materials processing facilities.
- The required overall station capacity (i.e., the number and size of containers) depends on the size and population density of the area served and the frequency of collection.

(ii) **Large transfer stations:** These are designed for heavy commercial use by private and municipal collection vehicles. The typical operational procedure for a larger station is as follows:

- when collection vehicles arrive at the site, they are checked in for billing, weighed and directed to the appropriate dumping area;

- collection vehicles travel to the dumping area and empty the wastes into a waiting trailer, a pit or a platform;
- after unloading, the collection vehicle leaves the site, and there is no need to weigh the departing vehicle, if its weight (empty) is known;
- Transfer vehicles are weighed either during or after loading. If weighed during loading, trailers can be more consistently loaded to just under maximum legal weights and this maximises payloads and minimises weight violations.

Route Optimization

Once the equipment and labor requirements are determined, collection routes must be laid out so both the collectors and equipments are used effectively. In general the layout of collection route involves a series of trial. There is no universal set of route that can be applied for all the situations. Some guidelines that should be taken into consideration when laying out routes are as follows.

1. Existing policies and regulation related to such items as the point of collection and frequency of collection must be identified.
2. Existing system characteristics such as crew, size and vehicle type must be coordinated.
3. Whenever possible route should be laid down so that they begin and end near main arterial roads using topographical and physical barriers as route boundaries.
4. In hilly areas, route should start at the top of the grade and proceed downhill as the vehicle is loaded.
5. Routes should be laid out so that the last container to be collected on the route is located nearest to the disposal site.
6. Waste generated at traffic congested location should be collected as early as possible.
7. Sources at which extremely large quantities should be serviced during the first part of the day.
8. Scattered pick up points where small quantities of solid waste generation that receive the same collection frequency should if possible be serviced during one trip or on the same day.

IMPLEMENTING COLLECTION AND TRANSFER SYSTEM

Implementing of collection and transfer system involves the following:

(i) **Finalising and implementing the system management plan:** For proper implementation of collection and transfer system, it is necessary to have clear organisational structures and management plans.

- The organisational structure should be simple.
- All workers in the department should clearly understand the department's mission and their roles.
- Through training, incentives and reinforcement by management, workers should be encouraged to be customer-oriented and team contributors.
- Feedback mechanisms must be introduced to help the crew review their performance and help managers monitoring the performance of crews, equipment, etc.
- It is also important to periodically review the management plans and structures, as implementation of collection services continues.

(ii) **Purchasing and managing equipment:**

- For purchasing equipment, most municipalities issue bid specifications. Detailed specifications include exact requirements for equipment sizes and capacities, power ratings, etc.
- Municipalities may either perform equipment maintenance themselves, contract with a local garage, or in some cases, contract with the vehicle vendor at the time of purchase.
- each vehicle should have an individual maintenance record that includes preventive maintenance schedule; current list of specific engine; a description of repairs and a list containing information on the repair date, mechanic, cost, type and manufacturer of repair parts and the length of time the truck was out of service, for each maintenance event.

(iii) **Hiring and training personnel:**

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- As in all organisations, good personnel management is essential to an efficient, high-quality waste collection system.
- The recruitment programme should assess applicants' abilities to perform the types of physical labour required for the collection, equipment and methods used.
- To retain employees, management should provide a safe working environment that emphasises career advancement, participatory problem solving and worker incentives.
- Feedback on employee performance should be regular and frequent.
- This programme should outline safety procedures and ensure that all personnel are properly trained on safety issues.
- Haulers should develop an employee training programme that helps employees improve and broaden the range of their job-related skills.
- Education should address such subjects as driving skills, first aid, safe lifting methods, identification of household hazardous wastes, avoidance of substance abuse and stress management.

(iv) **Providing public information:** Maintaining good communication with the public is important to a well-run collection system.

- Residents can greatly influence the performance of the collection system by co-operating in separation requirements, and by keeping undesirable materials from entering the collected waste stream.
- Commonly used methods of communicating information include brochures, articles in community newsletters, newspaper articles, announcements, and advertisements on radio and television, information attachments to utility bills (either printed or given separately) and school handouts.
- Communication materials should be used to help residents understand the community waste management challenges and the progress in meeting them.
- Residents should also be kept informed about issues such as the availability and costs of landfill capacity so that they develop an understanding of the issues and a desire to help meet their waste management needs.

(iv) **Monitoring system cost and performance:**

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Collection and transfer facilities should develop and maintain an effective system for cost and performance reporting. Each collection crew should complete a daily report containing the following information:

- Total quantity hauled.
- Total distance and travel times to and from the disposal site.
- Amounts delivered to each disposal, transfer, or processing facility.
- Waiting time at sites.
- Number of loads hauled.
- Vehicle or operational problems needing attention.

Collected data should be used to forecast workloads, truck costs, identify changes in the generation of wastes and recyclables, trace the origin of problematic waste materials and evaluate crew performance. A monitoring system provides the short-term feedback necessary to identify the corrections needed to achieve those goals.

THE CASE OF BANGALORE

In the Bangalore city (India), the waste collected through street sweeping is the main system of primary collection of wastes. However, recently efforts are being made for doorstep collection of waste through NGOs (Non-Governmental Organisations) and private contractors, but only about 5% of the population is covered under this system. The waste generated by the rest is collected from either the street or the dustbins. Other details regarding the collection process in Bangalore are given below:

(i) **Waste storage:** There are about 14,000 bottomless cement bins having 0.9 meters diameter and 0.6 cubic meter storage capacity and large masonry bins for depositing wastes at a distance of about 100 to 200 meters. Besides these, there are 1500 places, where the waste is deposited but no bins are kept on these sites. Recently, metal containers have been placed and at present 55 metal containers are in the city for the storage of waste in a more hygienic manner.

(ii) **Waste collection:** The frequency of removal of wastes varies from place to place, depending on the locality. Whichever system adapted in the area needs proper planning for collection, loading, unloading and transportation from transfer station and to the point of final disposal, considering traffic constraints, peak hour traffic, etc. An optimum collection schedule requires to be worked out where the number of premises or dumps is mentioned on a daily programme sheet, to be executed by the driver or supervisor in

charge of collection. At present, it is estimated that there are about 4943 hotels/restaurants, which produce a large quantity of organic wastes in Bangalore. The silt and waste removed from drains get deposited along the roadside. The human and animal excreta also add to the mass getting deposited on streets. Mechanical sweeping or cleaning cannot work in Bangalore roads and footways because of obstructions due to the activities of hawkers, shop extensions, broken pavements, etc. Pedestrians, shops, goods vehicles carrying loose materials contribute to street littering of paper, used tickets, cigarette butts, etc., as well as vehicles dropping material during their movement.

(iii) **Waste transportation:** Removal of garbage is a very important aspect of SWM, and the method of transportation is crucial. In essence, any breakdown in this system could create problems. Transportation implies conveyance from point of collection to the point of final disposal either directly or through a transfer system. In Bangalore, the transportation of waste is done by: engaging, departmentally, 82 trucks of the Corporation. engaging 129 vehicles, on contract, for layout and markets and 72 vehicles for transportation of waste. (In addition, the Bangalore Corporation has 13 dumper placers for transporting metallic containers of 2.5 to 3 tonnes capacity and 6 mini-compactors for transportation of waste.)

The clearing efficiency is 30 to 35%. The vehicles are open and the spillage of waste on the roads is a common feature. Each truck makes two trips to the dumpsite every day. There are no transfer stations in Bangalore. The waste collected from the roads and bins is directly transported to the final disposal sites. There is no arrangement made for the primary collection of construction waste. The engineering division of the Corporation removes the unauthorised construction waste from time to time. There are 115 small and big vegetable, fruit and meat markets in the city. However, no special arrangements are made for the collection of waste from these markets.

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A. Combined direct and storage discharge.

D. Direct discharge – In a direct discharge transfer station, wastes from the collection vehicles usually are emptied directly into the vehicle to be used to transport them to a place of final disposal.

To accomplish this,

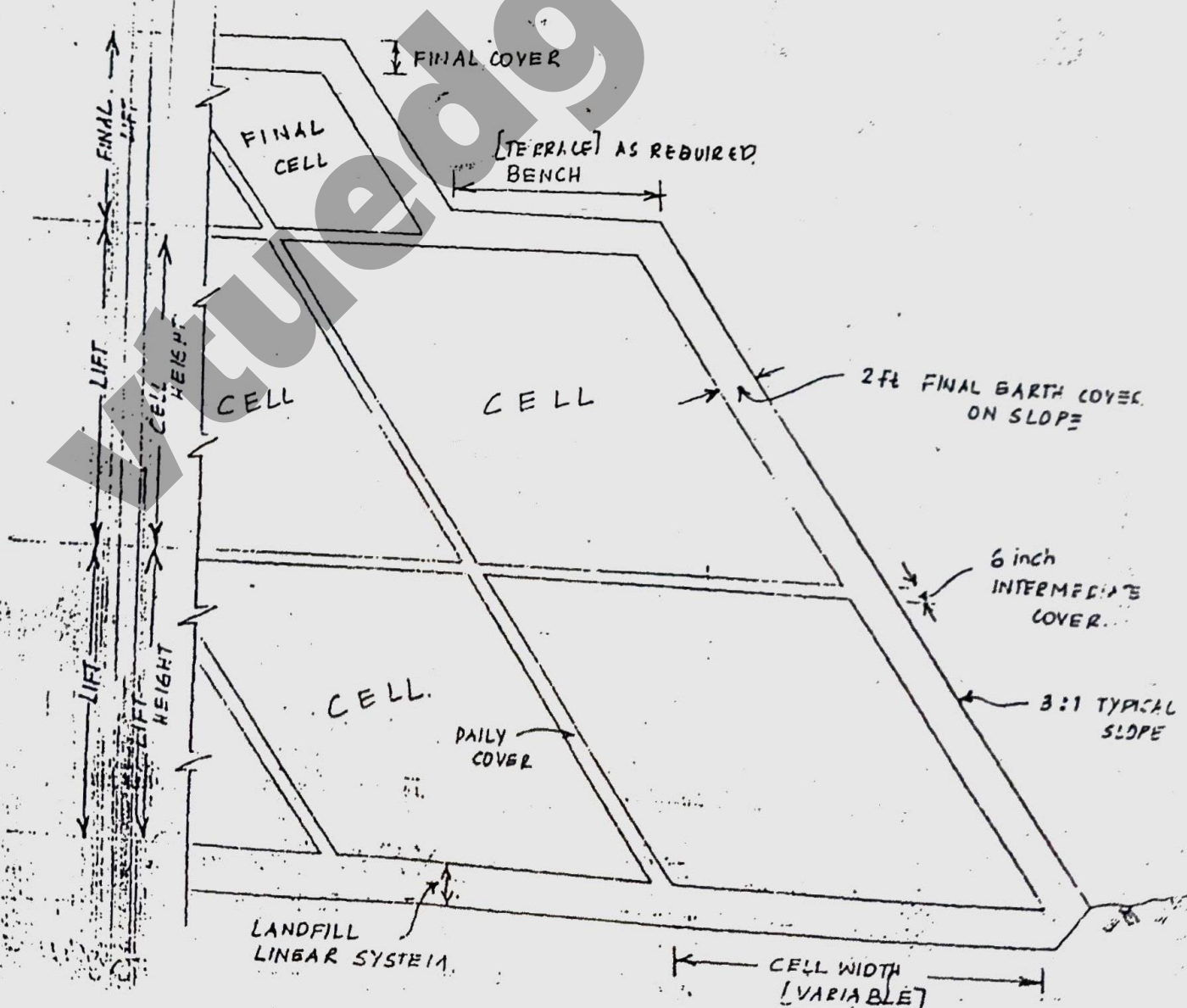
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SANITARY LANDFILLING

INTRODUCTION → Land fills have been the most economical and environmentally acceptable method for the disposal of solid waste. Even with implementation of waste reduction, recycling and transformation technologies, disposal of residual solid waste still remains an important component of an integrated solid waste management strategy. Landfill management incorporates planning, design, operation, closure and post closure control of land fills.

Based on the past experience throughout the world, land disposal in the form of sanitary landfill has proved to be most economical and acceptable method for the disposal of solid waste. OPEN DUMPING, as distinguished from sanitary landfilling is still used in parts of the country, but is no longer an acceptable means of land disposal from an Aesthetic, environmental and Sanitary standby.

SCHEMATIC VIEW THROUGH A SANITARY LANDFILL



DEFINITION OF TERMS

LANDFILLS

→ are the physical facilities used for the disposal of residual solid wastes in the surface soils of the earth.

SANITARY LANDFILL

→ An operation in which the waste to be disposed off are compacted and covered with a layer of soil at the end of each days operation. When the disposal site has reached it's ultimate capacity i.e. after all disposal operations have been completed, a final layer of 2ft @ more of cover material is applied.

Today sanitary landfill refers to an engineered facility for the disposal of MSW designed and operated to minimize public health and environmental impacts.

SECURE LANDFILLS

Landfills for the disposal of Hazardous wastes.

LAND FILLING ACTIVITY

→ is the process by which residual solid waste is placed in a landfill and it includes monitoring of incoming waste stream, placement and compaction of the waste and installation of landfill environmental monitoring and control facilities.

CELL

→ It is used to describe volume of material placed in a landfill during one operating period usually one day. It includes the solid waste deposited and also the daily cover material surrounding it.

DAILY COVER

→ Usually consists of 6-12 inches of native soil @ alternate materials such as compost that are applied to the working faces of the landfill at the end of each operating period. The purposes of daily cover are

- ① To control the blowing of waste materials.
- ② To prevent rats, flies and other

disease vectors from entering @ existing of the landfill, and.

(iii) To control the entry of water into the landfill during operation.

LIFT →

is a complete layer of cells over the active area of the landfill, typically landfills are comprised of ~~series~~ of lifts.

BENCH ON TERRACE →

is commonly used where the height of the landfill will exceed 50-75 ft. Benches are used to maintain the slope stability of the landfill, for the placement of surface water drainage channels and for the location of landfill gas recovery piping.

FINAL COVER LAYER →

It is applied to entire landfill surface after all land filling operations are complete. It usually consists of multiple layers of soil and/or Geomembrane materials designed to enhance surface drainage, intercept percolating water and support surface vegetation.

LEACHATE →

The liquid that collects at the bottom of a landfill is known as LEACHATE. In general leachate is a result of the percolation of precipitation, uncontrolled runoff and irrigation water into the landfill. Leachate can also include water initially contained in the waste as well as infiltrating ground water. Leachate contains a variety of chemical constituents derived from the solubilization of materials deposited in the landfill and from the products of chemical and bio-chemical reactions occurring within the landfill.

LANDFILL GAS →

Is the mixture of gases formed within a landfill. The bulk of landfill gas consists of Methane and CO_2 . The principal products of Anaerobic decomposition of biodegradable organic fraction of the MSW in the landfill. Other components of landfill gas include atmosphere

Nitrogen and oxygen, ammonia and trace organic compounds

LANDFILL LINERS

→ are the materials both natural and manufactured, that are used to line the bottom area and below grade sites of a landfill. Liners usually consists of layers of compacted clay and / or geomembrane materials designed to prevent migration of landfill leachate and landfill gas.

LANDFILL CONTROL FACILITIES

→ Includes liners, landfill leachate collection and extraction systems, landfill gas collection and extraction systems and daily and final cover layers.

LANDFILL CLOSURE

→ It is the term used to describe the steps that must be taken to close and secure a landfill site once the filling operation has been completed. Post-closure care refers to the activities associated with the long term monitoring and maintenance of the completed landfill [typically 30-50 years].

ADVANTAGES AND DISADVANTAGES OF SANITARY LANDFILL

ADVANTAGES	DISADVANTAGES
where land is available, a sanitary landfill is usually the most economical method of solid waste disposal. The initial investment is low compared with other disposal methods like incineration or pulverisation. A sanitary landfill is a complete final disposal method as compared to incineration and composting which require additional treatment or disposal operations for leachate, unusable	In highly populated areas suitable land may not be available within economical hauling distance. Proper sanitary landfill standards must be followed to avoid the operation may result in an open dump. Sanitary landfills located in residential areas can provoke extreme public opposition.

ADVANTAGES OF SANITARY LANDFILL:-

- Where land is available, a sanitary landfill is useful and proves to be most economical method of solid waste disposal method.
- The initial investment is low compared to other disposal methods.
- A sanitary landfill is a complete and final disposal method as compared to incineration and composting.
- A sanitary landfill can receive all types of solid waste, eliminating the necessity of separate collection.
- It is flexible, increased quantities can be handled.

- A sanitary landfill can receive all types of wastes, eliminating the necessity of separate collection.
- Sub marginal land may be reclaimed for use as parking, playgrounds, airports etc.

DISADVANTAGES OF SANITARY LANDFILL:-

- In highly populated areas suitable land may not be available within economical hauling.
- Proper sanitary landfill standards must be followed daily else it ends up to be a dumping site.
- Sanitary landfills located in residential areas can provoke extreme public opposition.
- A completed landfill will settle gradually and require periodic maintenance.
- Special design and construction required to be utilized for buildings constructed on completed landfills because of the settlement factor.
- Methane, an explosive gas and other gases produced from the decomposition of the waste may become a hazard or nuisance and interfere with the use of the completed landfill.
- Possibility of Soil & Ground water Contamination due to leachate.

SITE SELECTION FOR A LANDFILL:-

- 1) Available land area -
Site should be having a useful life of more than one year.
- 2) Haul distance -
The length of the hauling distance will have an effect on the design of landfill.
- 3) Soil conditions and Topography -
Cover materials must be available locally and near to the site.
- 4) Surface water hydrology -
It impacts the drainage requirements.
- 5) Climatological Conditions :-
Provision must be made for wet weather operation.
- 6) Geology and Hydrological Conditions -
This is the most important factors in a landfill site wrt site preparation.
- 7) Local environmental conditions:-
Noise, odour, aesthetic control requirements.
- 8) Ultimate use of site:-
Affects long term management of site.

9) Location restrictions:-

Restrictions apply w.r.t landfills near airports, flood plane, wet lands, areas of faults, seismic zones and in unstable areas.

10) Site access:-

As the number of operating landfills continue to decrease, new landfills that are increasing in size, because land areas of suitable size are not readily available near existing developed roadways & cities, construction of excess roadways & rail lines.

LANDFILLING METHODOLOGY :

PLACEMENT OF WASTE IN THE SANITARY LANDFILL

Once the land fill sites have been prepared, the next step in the process involves actual placement of waste materials. The waste is placed ~~in~~ at the beginning, along the compaction face continuing upwards and outwards of the face. The waste deposited in each operating period and [usually one day forms an individual cell] waste deposited by vehicles are spread out in 18-24 inch layers and compacted. Typical cell height vary from 8-12 feet. Length of working face varies with site conditions and size of operation. The working face is the area of the land fill where the solid waste is unloaded, placed and compacted during a given operating period. Width of the cell varies from 10-30 feet. All exposed faces of the cell are covered with thin layer of soil 6-12 inches thick at the end of each operating period. After one @ more lifts have been placed horizontal gas recovery trenches can be excavated in the completed surface. The trenches are filled with gravel and perforated plastic pipes are installed in trenches. Gas is extracted through pipes depending on the depth of landfill, additional leachate collection facilities may be placed in lifts. A cover layer is applied to the completed landfill section. The final cover is designed to minimize infiltration of leachate and to

also cover is landscaped to
vertical gas extraction systems may be installed
through completed landfill surface and the whole
system is tied together and routed to energy
recovery facilities. Reactions in sanitary landfills.

The various Reactions in sanitary landfilling are

- (i) Biological Reactions
- (ii) Chemical Reactions
- (iii) Physical Reactions.

BIOLOGICAL REACTIONS → The most important biological reactions that occur in landfills are those involving organic matter in MSW that led to the development of landfill gases and liquids. The biological decomposition process usually proceeds AEROBICALLY for some short period immediately after deposition of waste until the oxygen initially present is depleted. During Aerobic decomposition CO_2 is the principle gas produced. Once all the available oxygen has been consumed, the decomposition becomes Anaerobic and the organic matter is converted to CH_2 and CH_4 and trace amounts of Ammonia, H_2S , many other chemical reactions are biologically mediated as well. Because of number of inter-related influences, it is difficult to define the reactions that will exist within the landfill.

CHEMICAL REACTIONS → Important chemical reactions that occurs within the landfill include dissolution and suspension of material and biological conversion products in the liquid percolating through waste evaporation and vapourisation of chemical compounds and water into evolving landfill gas. Sorption of volatile and semi volatile organic compounds [OC's] into the landfilled materials, de-halogenation and decomposition of organic compounds and oxidation-reduction reactions affecting metals and solubility of metal salts. The dissolution of biological conversion products and other compounds, particularly of organic compounds into leachate is of special materials. because these materials can be transported out of the leachate. These organic compounds can be subsequently released into the atmosphere either through soil or from uncovered leachate recovery facilities. Other important chemical reactions include those between

certain organic compounds and clay liners which may alter the structures and permeability of the liner material.

PHYSICAL REACTIONS

→ Among the more important physical changes in landfills are, the lateral diffusion of gases in landfill and emission of landfill gases to the surrounding environment, movement of leachate within the soil and into the landfill and into the underlying soil, and settlement caused by decomposition of landfilled materials. Landfill gas movements and emission are very important in management as gas is evolved within a landfill, Internal pressure may build causing landfill to crack and leak. Though the water may enter into the landfill and enhance the gas production rate causing still more cracking and escaping landfill gases may carry trace carcinogenic substances into the surrounding environment. Because landfill gas usually has a high methane content, there may be a combustion/explosion hazard. Leachate migration is another important concern, it may transfer and transport the dangerous materials to new location.

SANITARY LANDFILLING METHODS

The principal methods used for the landfilling of MSW are

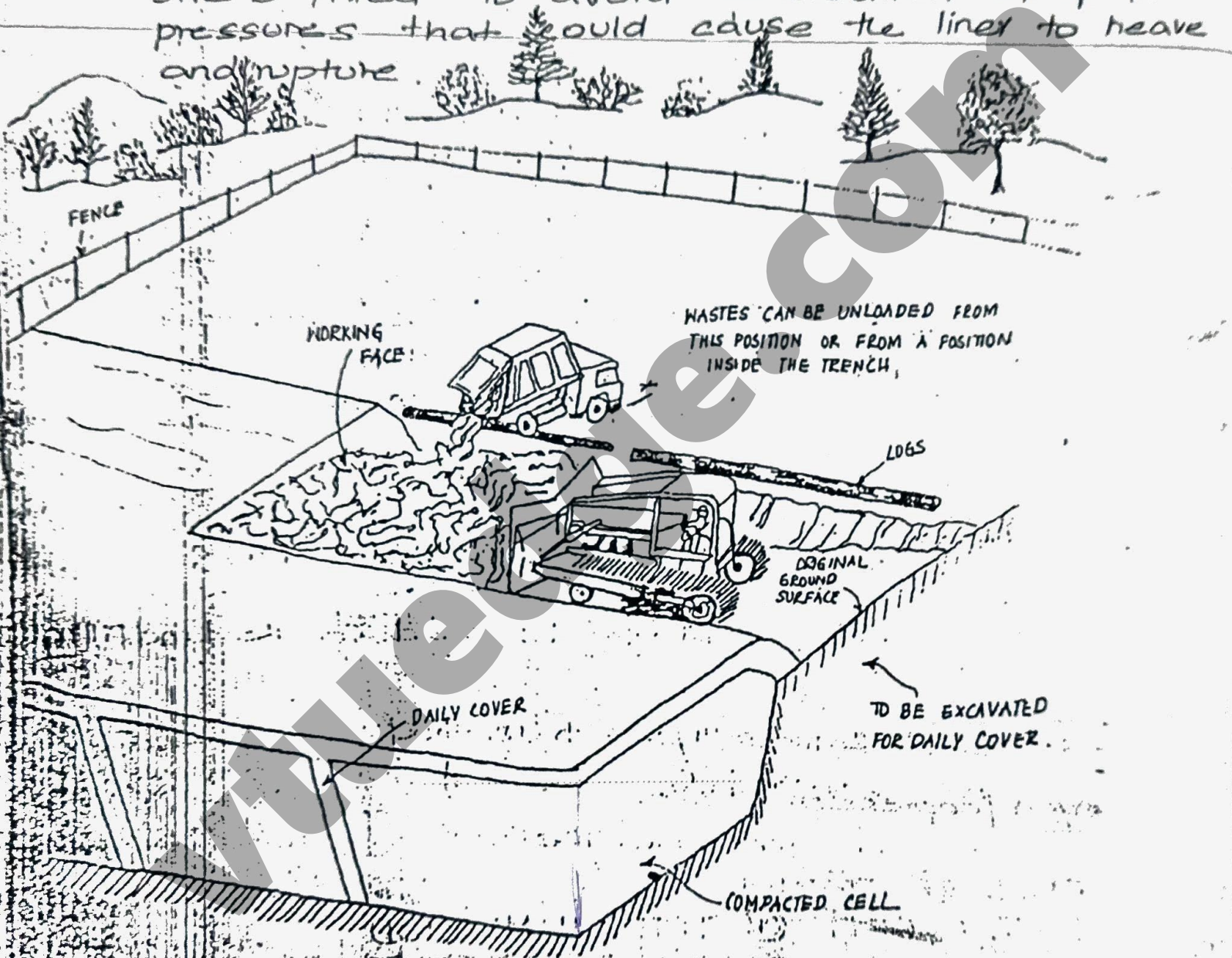
- (i) excavated cell/trench method
- (ii) Area method (Ramp method)
- (iii) Canyon/depression method

EXCAVATED CELL OR TRENCH METHOD

The excavated cell/trench method of landfilling is ideally suited through areas where an adequate depth of cover material is available at the site and where the water table is not near the surface. Typically S.W. are placed in cells or trenches excavated in the soil. The soil excavated from the site is used for daily and final cover. The excavated cells/trenches are usually lined with synthetic membrane liners or low permeability clay or a combination of the two to limit the movement of

landfill gases and leachate. Excavated cell are typically square, upto 1000 ft in width and length, with side slopes of 1.5:1 to 2:1 in length, 3-10 ft in depth, and 15-50 ft in width.

Landfills are allowed to construct below the high ground water level if special provisions are made to prevent ground water from entering the landfill and to contain @ eliminate the movement of leachate @ gases from completed cells. Usually the site is dewatered, excavated and then lined in compliance with local regulation. The dewatering facilities are operated until the site is filled to avoid the creation of uplift pressures that could cause the liner to heave and rupture.

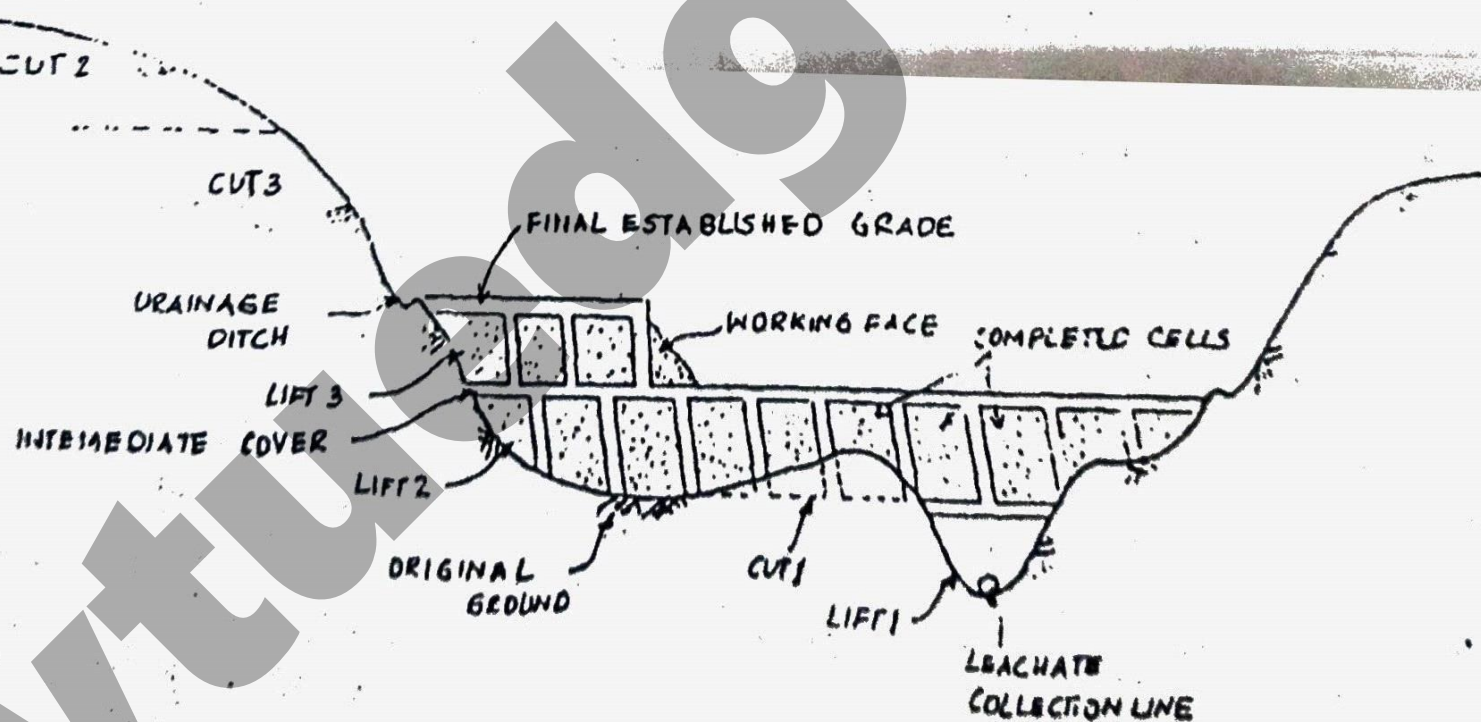


AREA METHOD [RAMP METHOD]

It is used when the terrain is unsuitable for the excavation of cells @ trenches in which to place the S.W. In high ground water conditions necessitate the use of Area Method. Landfill site preparation include the installation of liner and leachate control system. Cover material must be hauled in by truck and earth moving equipment from adjacent land @ from borrow pit areas. In locations with limited availability of materials that can be used as cover, compost produced from

yard waste and M.S.W has been used successfully as intermediate cover material. Other techniques that have been used include the use of movable temporary cover materials such as soil and geo-membranes. Soil and Geo-membranes placed temporarily over a completed cell can be removed before the next lift is begun.

CANYON @ RAVINE



CANYONS, RAVINES, DRY BORROWPITS and QUARRIES have been used for land fills. The Techniques to place and compact solid waste in canyons @ depression landfills vary with

- (i) Geometry of site
- (ii) characteristics of available cover material
- (iii) Hydrology and Geology of site
- (iv) Type of leachate and gas control facilities to be used
- (v) Access to the site

In a CANYON site filling starts at the head [higher] end of the canyon and ends at the mouth. This practice prevents accumulation of water behind landfill. Waste usually are deposited on the canyon floor and from there are pushed against the canyon face at a slope of 2:1. In this way a high degree of compaction can be achieved.

Control of surface drainage often is a critical factor. A key to the successful use of this method is the availability of adequate material to cover the individual lift as they are completed and to provide the entire landfill when the final height is reached. Cover material is excavated from the canyon walls @ floor before the liner system is installed. Borrowpits and abandoned quarries may not contain sufficient soil for intermediate cover so that cover material may have to be imported. Compost produced from yard waste and MSW, can also be used for intermediate cover layers.

GASES AND LEACHATE

composition, characteristics, generation & movement of land filled gases.

COMPOSITION

→ Land fill gas is composed of Number of gases that are present in large amounts (principal gases) and No. of gases present in small amounts (trace gases). The principal gases are produced by the decomposition of MSW. Some of the trace although present in small quantities may be toxic and could present risk to public health.

PRINCIPAL LANDFILL GAS CONSTITUENTS

Gases found in landfill includes NH_3 , CO_2 , CO , H_2S , CH_4 , N_2 , H_2 and O_2 .

CH_4 and CO_2 are the principal gases produced from ANAEROBIC decomposition of biodegradable organic components in MSW. When CH_4 is present in air in concentration b/n 5-15%, it is explosive. Because only limited amount of oxygen is present in a landfill when CH_4 concentration reach this critical level. There is little danger that the landfill will explode. However CH_4 mixtures in explosive range can form if landfill gases migrates offsite and mixes with air. The concentration of these gases that may be expected in the leachate will depend on their concentration in the gas phase in contact with the leachate.

TRACE LANDFILL GAS CONSTITUENT

In a study conducted in England, gas samples were collected from three different landfills and analyzed were 154 compounds. A total of 116 compounds were found in landfill gas, many compounds could be classified as Volatile organic compounds (VOC).

The trace gases found in landfill includes Acetone, Benzene, chloro benzene, chloroform, Di-chloro methane, Ethylene bromide, Toulene, Tetrachloro-ethylene, Vinyl Acetate, Xylene, Vinyly chloride, etc.....

The occurrence of significant concentration of VOC is associated with older landfills that accepted industrial and commercial waste containing VOC. In newer landfills in which the disposal of hazardous waste is banned, the concentration of VOC in landfill gas have been extremely low.

ACCURANCES OF GASES AND LEACHATE IN LANDFILLS

The following biological, physical and chemical events occur when solid waste are placed in sanitary landfill.

- ① Biological decay of organic materials either Aerobically or Anaerobically with evolution of gases and liquids.

- (ii) Chemical oxidation of waste materials.
- (iii) Escape of gases from the landfill.
- (iv) Movements of gases caused by differential heads.
- (v) Dissolving and leaching of organic and inorganic materials by water and leachate moving through the landfills.
- (vi) Movement of dissolved materials from concentrated gradient and osmosis.
- (vii) Uneven settlement caused by consolidation, settlement into voids.

GAS MOVEMENT

Under ideal conditions, the gases generated from a landfill should be either vented to the atmosphere (i) in larger landfills collected for the production of energy. In most cases over 90% gas volume produced from decomposition of solid waste and of CH_4 and CO_2 . Although most of the CH_4 escapes to the atmosphere, both CH_4 and CO_2 have been found in concentration upto 40% at lateral distances of upto 130m at the edges of the landfill. If vented into the atmosphere in an uncontrolled manner CH_4 can accumulate [because its specific gravity is less than air] below the buildings (ii) other enclosed spaces are closed to a sanitary landfill with proper venting, CH_4 should not be a problem.

CO_2 is about 1.5 times as dense as air and 2.0 times as dense as CH_4 , it tends to move towards the bottom of the landfill. As a result the concentration of CO_2 in the landfill may be high for years. Ultimately because of this density CO_2 will also move downwards through the underlying formation until it reaches ground water. Because CO_2 is readily soluble in H_2O , it usually lowers the pH value which in turn can increase hardness and mineral content of ground water through stabilization [solubilization] of calcium and magnesium carbonates.

COMPOSITION, FORMATION, MOVEMENT AND CONTROL OF LEACHATE IN LANDFILL

may be defined as liquid that has percolated through solid waste and has extracted dissolved and suspended materials and is composed of the liquid that has entered the landfill from external sources such as surface drainage, rainfall, ground water and water from underground springs and liquid produced from decomposition of solid waste.

COMPOSITION OF LEACHATE → When water percolates through S.W. that are undergoing decomposition both biological materials and chemical constituents are leach into the solution. The chemical composition of leachate will vary greatly depending upon the age of the landfill and events preceeding the type of sample.

Eg:- If a leachate sample is collected during acid phase of decomposition PH value will be very low. The composition of BODs, TOC, COD, Nutrients and heavy metals will be high. On the other hand leachate sample that is collected during methane fermentation phase, the PH value will be in the range of 6.5 - 7.5 and BOD, COD, TOC and Nutrient values will be lower. Similarly concentration of heavy metals will be lower because most metals are less soluble at Neutral PH values of the leachate and will depend not only on the concentration of acids that are present but also on partial pressure of CO_2 in landfill gas that is in contact with leachate. The biodegradability of leachate will vary with time, changes in biodegradability of leachate will vary with time. Changes in biodegradability can be monitored by checking BODs and COD ratios. Initially the ratios will be in the range of 0.5 @ greater; ratios in the range of 0.4 - 0.6 are taken as indicators that the organic material is readily degradable [1:1.6 to 2.6].

In matured landfills this ratio is often in the range 0.05 to 0.2, the ratio drops because the leachate from matured landfills typical contains humic and fulvic acids, which are readily not bio-degradable. As a result of variability in leachate characteristics, the design of leachate treatment facilities are complicated. Usually the treatment of the leachate will be taken care of domestic waste water treatment plants.

LEACHATE MOVEMENT

Under normal conditions, leachate is found in the bottom of landfills. From there, its movement is through the underlying strata, although some lateral movement may also occur, depending on the characteristics of the surrounding material. The rate of seepage of leachate from the bottom of a land fill can be estimated by Darcy's law by assuming that the material below the landfill to the top of the water table is saturated and that a small layer of leachate exists at the bottom of the fill. Under these conditions the leachate discharge rate per unit area is equal to the value of the co-efficient of permeability 'K' expressed in meters per day. The computed value represents the maximum amount of seepage that would be expected, and this value should be used for design purposes. Under normal conditions, the actual rate would be less than this value because the soil column below the landfill would not be saturated.

CONTROL OF LEACHATE MOVEMENT

As leachate percolates through the underlying strata, many of the chemical and biological constituents originally contained in it will be removed by the filtering and adsorptive action of the material composing the strata. In general, the extent of this action depends on the characteristics of the soil, especially the clay content. Because of the potential risk involved in allowing leachate to percolate to the ground water, best practice calls for its elimination or containment. Ultimately, it may be necessary to collect and treat the leachate.

The use of clay has been the favored method of reducing @ eliminating the percolation of leachate. Membrane liners have also been used; but they are expensive and require care so that they will not be damaged during the filling operations. Equally important in controlling the movement of leachate is the elimination of surface-water infiltration, which is the major contributor to the total volume of leachate. With the use of an impermeable clay layer, and appropriate surface slope [1 to 2 percent] and adequate drainage, surface infiltration can be controlled effectively. Generalized ratings for the suitability of various types of soil for use as a landfill cover

DISPOSAL METHODS

OPEN DUMPING

In this Method the refuse is collected, conveyed and openly dumped into low lying areas of the locality. The refuse should preferably consist of rubbish, Ashes, street washings and exclude Garbage. It is finally covered with a blanket of Earth @ Ash to avoid the nuisance of flies and Vermins. The Indian refuse is of heavier type and this method is found to be most convenient for our country. It is also adopted for rural communities, Hostels, camps, etc.

ADVANTAGES

- (i) This is a cheap method for a place where low lying land is easily available for this purpose.
- (ii) It does not require separation of different kinds of refuses as is necessary in other methods.
- (iii) It is most simple and economical.
- (iv) In low lying waterlogged areas and old quarry pits can be easily replanned by this method.
- (v) There are no residues @ by products left out. @ evolved in this method and hence the question of further disposal does not arise.

DISADVANTAGES

- (i) It becomes necessary to use insecticides for preventing the fly nuisance.
- (ii) The highly poisonous leachate that is formed during the rainy season due to the seepage of water through the area may pollute the underground sources of water supply.
- (iii) The low lying @ dumping sites may not always be available nearby.
- (iv) There is a continuous evolution of foul gases near the site and the gases formed may sometimes become explosive in nature.

FACTORS AFFECTING THE SELECTION OF SITE OF OPEN DUMPING.

- (i) The site should be away from the residential site.
- (ii) The site should be in a low lying area.

- ③ The source of water supply should not be near the open dumping pit.
- ④ The site should be easily accessible by Road
- ⑤ The site should not be located in an aquifer.

OCEAN DISPOSAL

This Method may be used to dispose off refuse by throwing it away into the sea, after carrying it to a reasonable distance from the coast (say 16-20 kms). The sea depth at such disposal point should not be less than 30 mts and the direction of currents should be such as not to bring it back towards the shore. This method may have a limited use in a few coastal towns.

DISADVANTAGES

- ① The Bulking and lighter parts of the refuse do not settle down, remain floating and tend to the shores especially during high tides.
- ② This Method requires ships and barges to take the refuse into the interior of the sea, the movement of which may be difficult during monsoons and stormy weathers. During such periods the refuse will either have to be collected or disposed off by some other methods.
- ③ Even inspite of the best care and effects, some refuse returns to the shores spoiling their beauty and hence this method is nowadays generally not used and has become obsolete.

ADVANTAGES

- ① The method is quite cheap and simple
- ② By the use of simple management technique, the method can be made as an environmental friendly disposal method of solid waste.
- ③ Large quantities of solid waste can be dumped into the oceans without use of costly equipments and skilled persons.

FEEDING TO HOGS

In this method the Garbage is fed to the animals like hogs, swines, etc... In Western countries and America this method is common. Garbage is the main element of refuse which is difficult to handle in sanitary conditions. In foreign countries the health authorities recommended heating of Garbage upto 100°C upto 30 minutes as a precaution to kill all the disease causing organisms.

DISADVANTAGES

- (i) It is responsible for disease and health hazards.
- (ii) ~~The raw feeding of Garbage to animals is responsible for diseases of Vesicular exanthema in swines and cholera among hogs.~~
- (iii) Animal feeding farms are the source of nuisance due to bad odours, flies and health hazards.
- (iv) It does ~~not~~ require separation of different kinds of refuse.

ADVANTAGES

- (i) It is most simple and economical.
- (ii) With proper precautions the swines and hogs can be fed at a cheaper rate.
- (iii) The Method is suitable for under developed countries, since it does not require any costly vehicle and equipments, and also technically qualified manpower.

Problem

- ① Determine landfill area required for a municipality with a population 50000, given that
- ① Solid waste generation = 350gms/person/day
 - ② Compacted density of landfill = 504 kg/m³
 - ③ Average depth of compacted S.W = 3

SOLUTION

I Calculation of total amount of S.W. generation in a day

$$\begin{aligned}\text{Total amount of S.W. generated in a day} &= \text{population} \times \text{per-capita S.W. generation} \\ &= 50000 \times 350 \\ &= 17.5 \times 10^6 \text{ gm/day} \\ &= 17.5 \times 10^3 \text{ kg/day}\end{aligned}$$

II Calculation of Volume of S.W

$$\begin{aligned}\text{Volume of S.W} = V &= \frac{\text{Total amount of S.W}}{\text{compacted density}} \\ &= \frac{17500 \text{ kg/day}}{504 \text{ kg/m}^3} \\ V &= 34.72 \text{ m}^3/\text{day}\end{aligned}$$

III Calculation of landfill area

Given that the Average depth of compacted S.W. is 3m.

$$\begin{aligned}\therefore \text{Total landfill area required for the Municipality} &= \frac{34.72}{3}\end{aligned}$$

$$\text{Total landfill area} = 11.57 \text{ m}^2$$

BENGALURU- A CASE STUDY ON LANDFIL:

The city has struggled with not being able to handle the load of the waste produced while employing a suitable system that integrates the existing informal waste disposal sector.

The rising population has contributed to enormous amounts of unrestricted new development which has in turn led to the expansion of the city in all directions. Bangalore's physical and economic growth have come with many challenges including rapid unplanned urbanization, flooding, urban sprawl, and an exponential increase in the amount of waste generated by the city, which has added to the city's woes.

From 2000 to 2022, the central pollution board of India reported that number of tons of garbage generated per day increased by an alarming 1750%. Second after Delhi, Bangalore saw the highest increase in garbage generation among other major cities in India, as it went from generating 200 tons per day in 2000 to 3,700 tons per day in 2022. More recent estimates are closer to 4,000 tons generated per day.

- **Disposal practices in Bengaluru**

Presently, Bengaluru does not have any scientific treatment method facilities for solid waste generated by municipal and industries around Bengaluru. This has led to the development of several illegal and unauthorized dump sites in Bengaluru. The waste produced by the bulk generators such as hotels, restaurants, kalayana mandapas, markets, etc., is being directly collected and transported to the treatment/disposal facilities. The treatment facilities have been developed around the city, and their spread over the entire zone

The closure of one of bangalore's biggest landfills at mandur, on the outskirts of the city, due to protests from villages that were nearby to the landfill. Bangalore employs a centralized waste disposal system; unsegregated waste that is collected is transported to landfills. Approximately 30% of the city's waste is collected by the bbmp, while the management of the remaining 70% is outsourced to contractors—from collection to disposal.

Sl.No	Zone	Existing Disposal Site/Facility
1	South	Bingipura, Mavallipura, KCDC
2	East	MSGP, Mavallipura, KCDC,
3	West	Terra firma, Mavallipura, KCDC
4	Yelahanka	Mavallipura, Terrafirma
5	Bommanahalli	Bingipura, Laxmipura
6	Mahadevapura	Terrafirma
7	Dasarahalli	MSGP
8	Raja Rajeshwari Nagar	MSGP/Terrafirma

- **Sanitary landfill site**

At present, Bengaluru can handle the municipal solid waste of about 2100 tpd only. The existing capacity of a waste treatment facility at mavallipura is 600 tpd, karnataka composting development corporation ltd (kcdc) is 300 tpd, and terra firma is about 1500 tpd. To achieve 100% processing of waste in Bengaluru, government has sanctioned six new facilities. These new facilities are being set up at the following locations: kannahalli is 500 tpd, seegihalli is 200 tpd, doddabidarakallu is 200 tpd, lingaderenahalli is 200 tpd, subrayanpalya is 200 tpd, chikkanagamangala is 500 tpd, kcdc (upgradation) is 500 tpd. These are majorly wet waste composting plants with a provision to screen compost out of mixed msw & provision also to store the non-compostable/ non-recyclable materials. These materials can be used for co-incineration in cement industries/ power generation.

- **Illegal dump sites**

There are more than 60 dump sites and it consists of both municipal and industrial waste existing in and around Bengaluru city. While the Bruhat Bengaluru Mahanagara Palike (BBMP) and the Karnataka State Pollution Control Board (KSPCB) close these sites, the new ones emerge elsewhere posing health risks to residents in their vicinity. The BBMP is merely collecting waste and dispose it. While the waste should be ideally segregated at the source, it is not done scientifically at present.

- **Transportation**

Transportation of solid waste from collection centers to final disposal site/landfill is another important step in the solid waste management. Currently, transportation of municipal solid waste is using pushcarts, auto, etc. Which bring the municipal solid waste to primary collection centers. From there trucks collect the municipal solid waste and transport it to waste disposal sites/landfill.